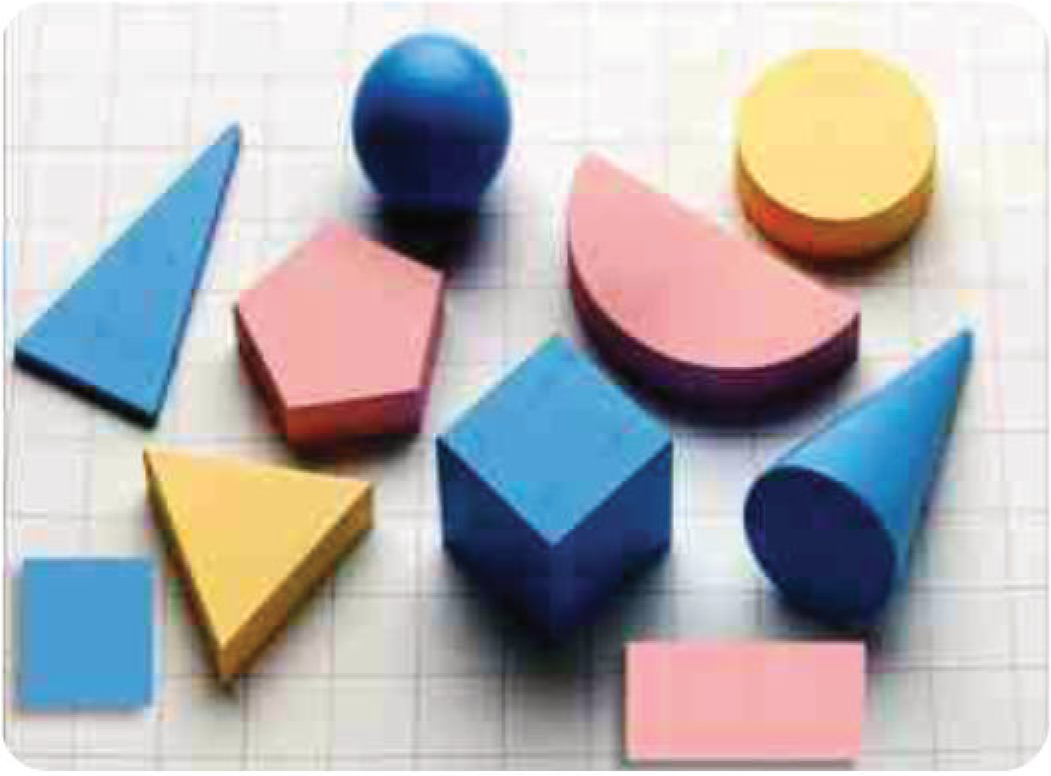
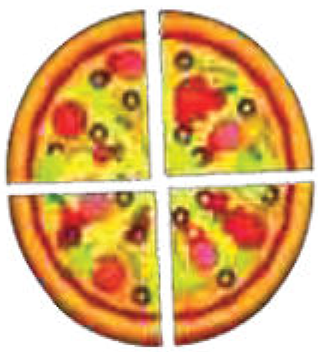
Geometry is all around us! Have you ever looked at a building, a toy, or even a puzzle and wondered about its shape? Geometry is the branch of mathematics that deals with shapes, sizes, and positions of objects.



Two Dimensional Shapes

What are 2-dimensional shapes?

Two-dimensional (2-D) shapes are flat shapes that have only length and width. We can find 2-D shapes all around us, from the shapes of objects to the patterns on the floor.



We learn each of them in detail.

Rectangle:

Look at these figures!

Mobile

Door

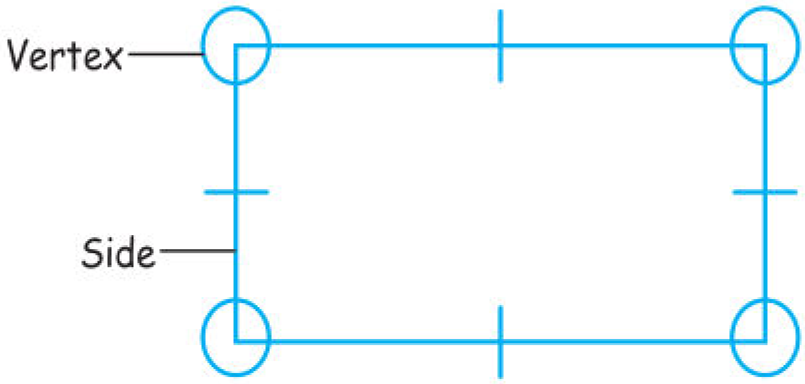
Book

Wooden block

All these figures are rectangular in shape.

Definition of Rectangle:

A rectangle is a two-dimensional shape with four corners (vertices) and four sides (edges).



- It has opposite sides of equal length, and all sides are straight.
- The longer side of the rectangle is its length, while the shorter side is its breadth or width.

To draw a rectangle, we draw four straight sides where opposite sides are equal in length.

Example:

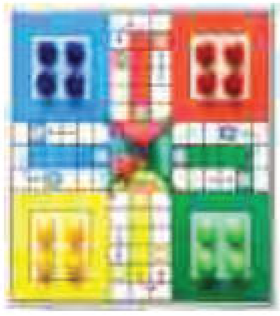
Draw a rectangle where its length is 10 centimeter and its breadth is 4 centimeters.

Square

Look at these figures!

Clock

LCD

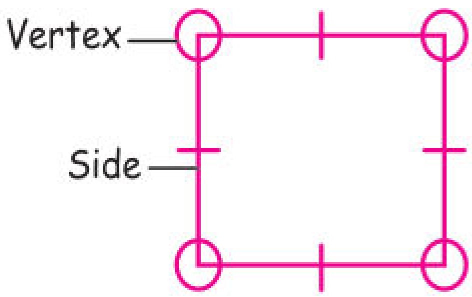


Ludo

Register

All these figures are square in shapes.

A square is a two-dimensional shape with four corners called vertices. All sides are equal in length and all sides are straight.

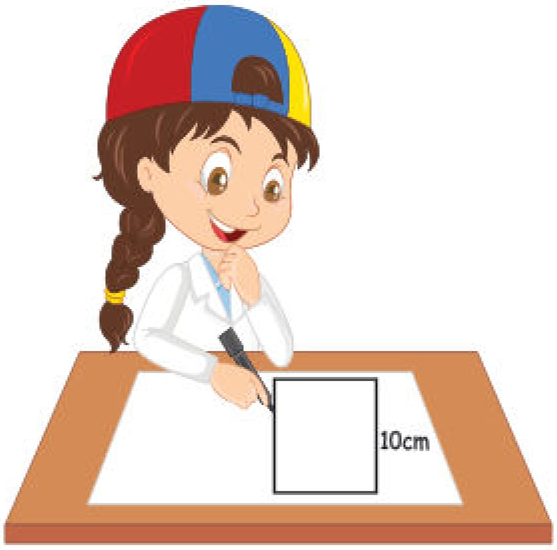


A square is a two-dimensional shape with four corners called vertices. All sides are equal in length and all sides are straight.

To draw a square, we draw four straight sides where all sides are equal in length.

Example:

Draw a square where length of each side is 10 centimeters.



Circle

Look at these figures!

Pizza

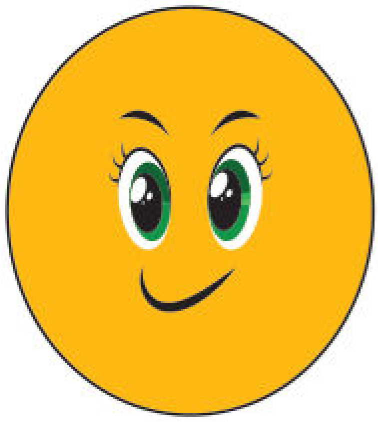
Clock

Ring

Wooden circular block

All these figures are circular in shape.

Circle



Definition of Circle:

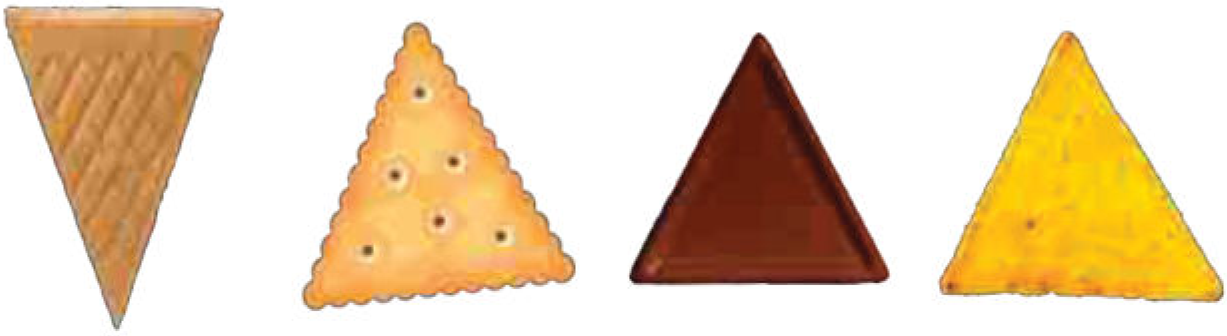
A circle is a round-shaped figure that has no corners or edges.

To draw the circle:

1. Make a small dot on your paper where you want the circle to be.
2. Place the tip of your pencil on the dot.
3. Move your pencil around the dot in a round and round motion, like you're making a big hug for the dot!

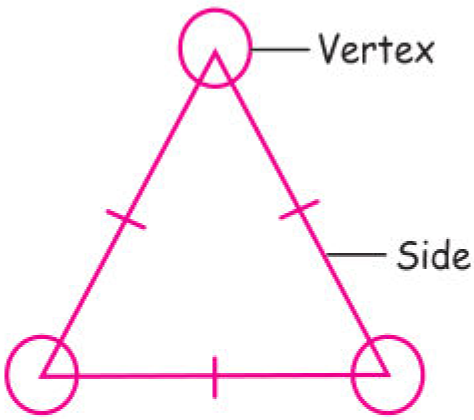
Triangle

Look at these figures!



All these figures are in triangular in shapes.

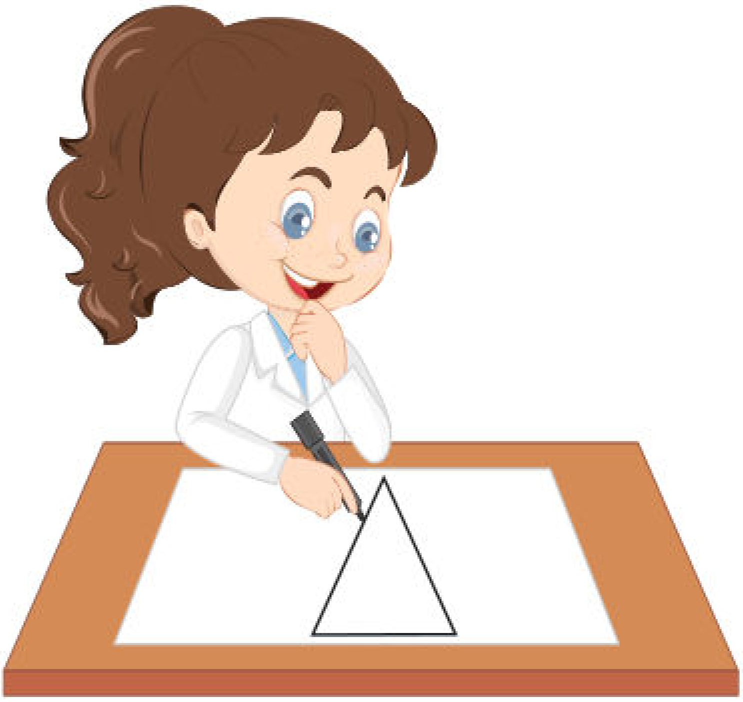
A triangle is a 2 dimensional shape with three edges (sides) having three (corners) vertices.



To draw the triangle:

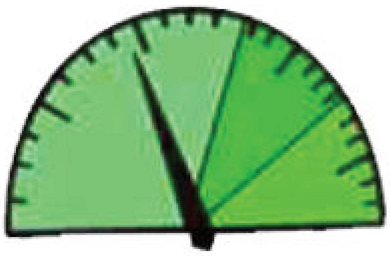
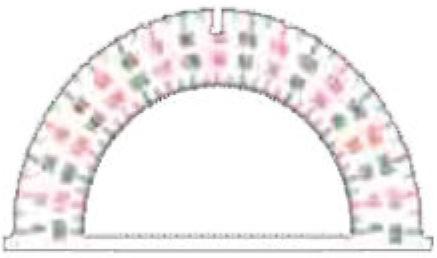
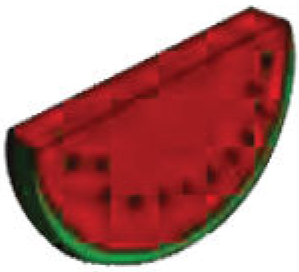
1. Start by drawing three dots on your paper, spaced evenly apart. These will be the corners of your triangle.
2. Connect the first dot to the second dot with a straight line.

3. Connect the second dot to the third dot with a straight line.
4. Finally, connect the third dot to the first dot with a straight line.



Semi-Circle

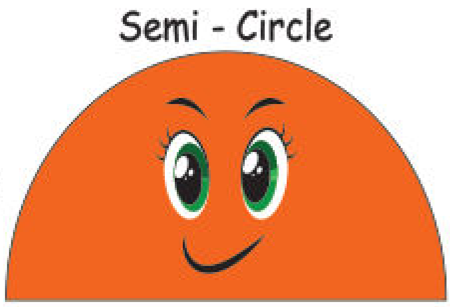
Look at these figures!

All these semicircular in shapes.

Definition of Semi-Circle

A semicircle is defined as a half circle formed by cutting the circle into two halves.

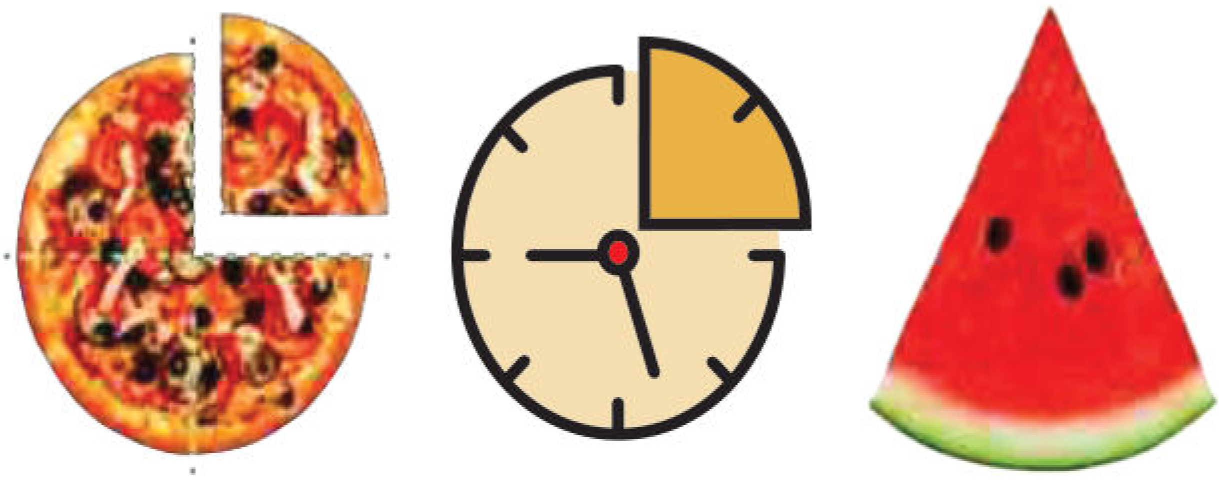


To draw the semi-circle,

1. Start by drawing a circle.
2. Draw a line through the circle, dividing it into two equal halves.

Quarter Circle

Look at these figures!

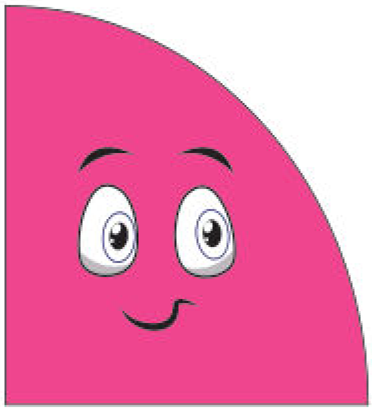


All these figures are quarter circular in shapes.

Definition of Quarter-Circle

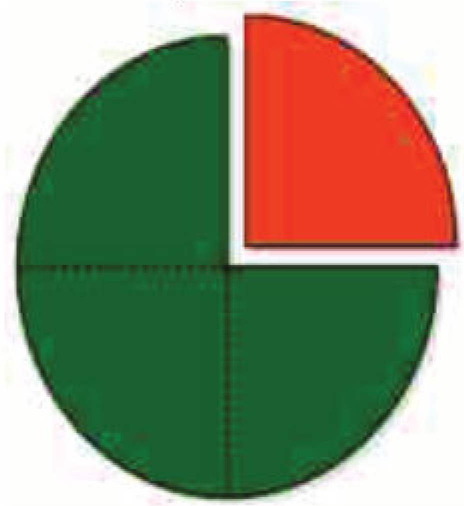
A quarter circle is a shape that is made by dividing a circle into four equal parts.

Quarter - Circle



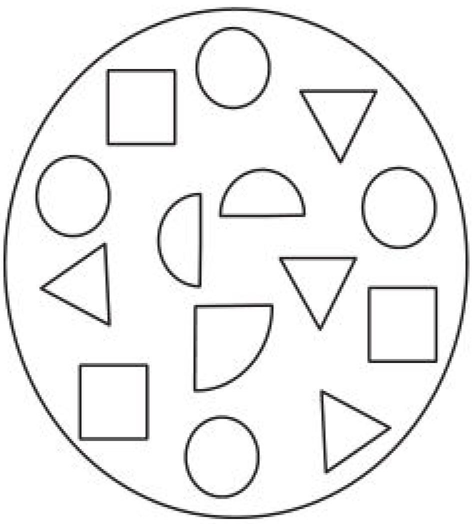
Drawing of Quarter Circle

1. Start by drawing a big circle.
2. Draw a line through the circle, dividing it into two halves.
3. Draw another line, dividing one of the halves into two quarters.

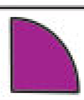
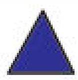
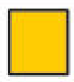
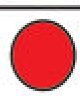


Exercise 1

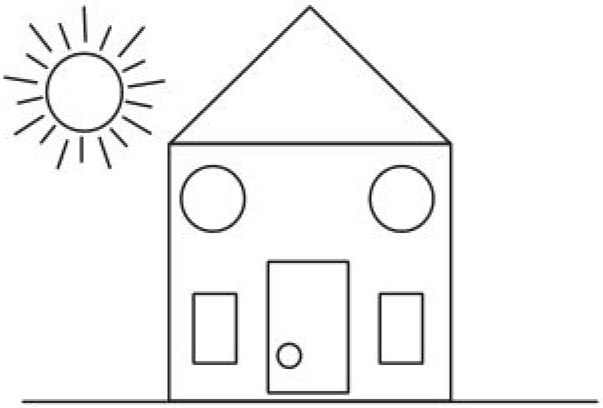
1. Colour the shapes according to the key provided.



Key:

Purple		Quarter circle
Blue		Triangle
Green		Semi-circle
Yellow		Square
Red		Circle

2. Colour the shapes according to the key provided.



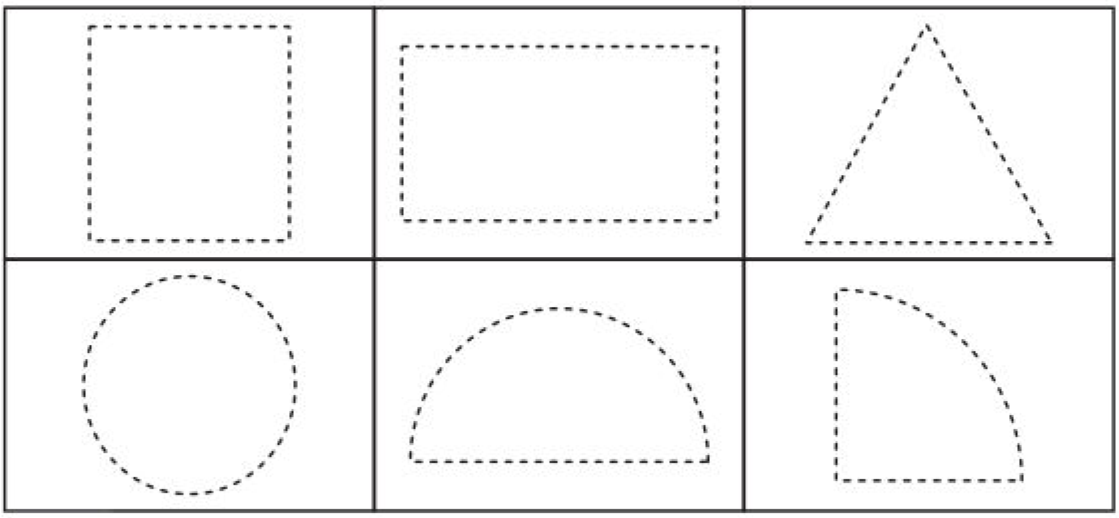
Key:

	Red
	Blue
	Green
	Yellow

3. Identify and label each shape. Draw related real life objects.

Circle			

4. Draw the following 2-D shapes

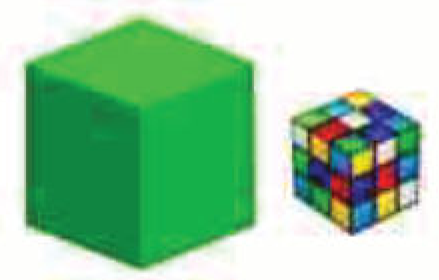


3-D shapes

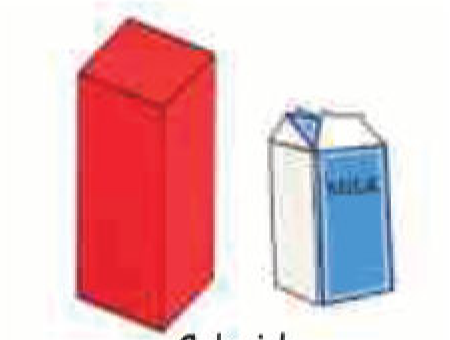
In our everyday life, we see many objects around us that have different shapes. Some objects have flat surfaces, while others are curved or rounded. In this section, we will learn about 3-D shapes that we see in our daily life.

Recognizing 3-D Shapes

A 3-D shape is an object that has length, width, and height. We can see 3-D shapes from different angles and orientations. Let us learn about five common 3-D shapes:



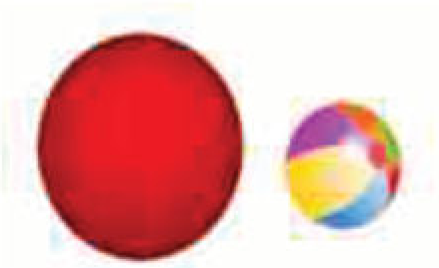
Cube



Cuboid

Cone

Cylinder

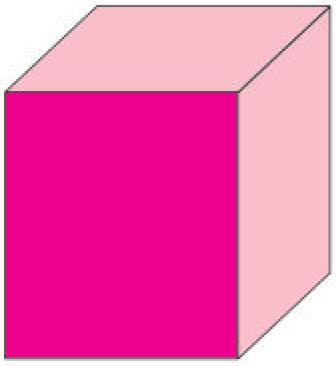


Sphere

Cube:

A cube is a shape with six square faces of equal size.

Cube

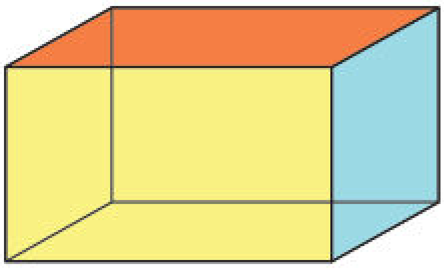


Example: A dice, a gift, a box and a dresser.

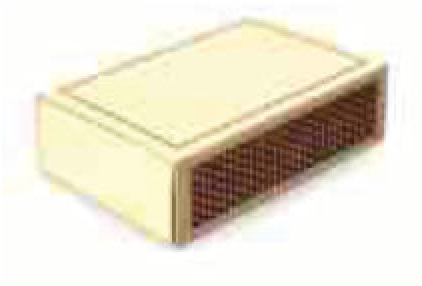
1. Cuboid:

A cuboid is a shape with six rectangular faces.

Cuboid



Example: A book, a brick and a match box.



Cone:

A cone is a shape with a circular base and a curved surface that tapers to a point.

Cone

Example: An ice cream cone, a party hat and a funnel

Cylinder:

A cylinder is a shape with two circular bases connected by a curved surface.

Cylinder

Example: A water bottle, a gas cylinder and a tin of soft drink

Sphere:

A sphere is a shape that is round and curved, like a ball.

Example: A basketball, a globe and an orange

Describing Position:

Objects can be found in different positions

Here are some examples of using these words to describe the position of objects:

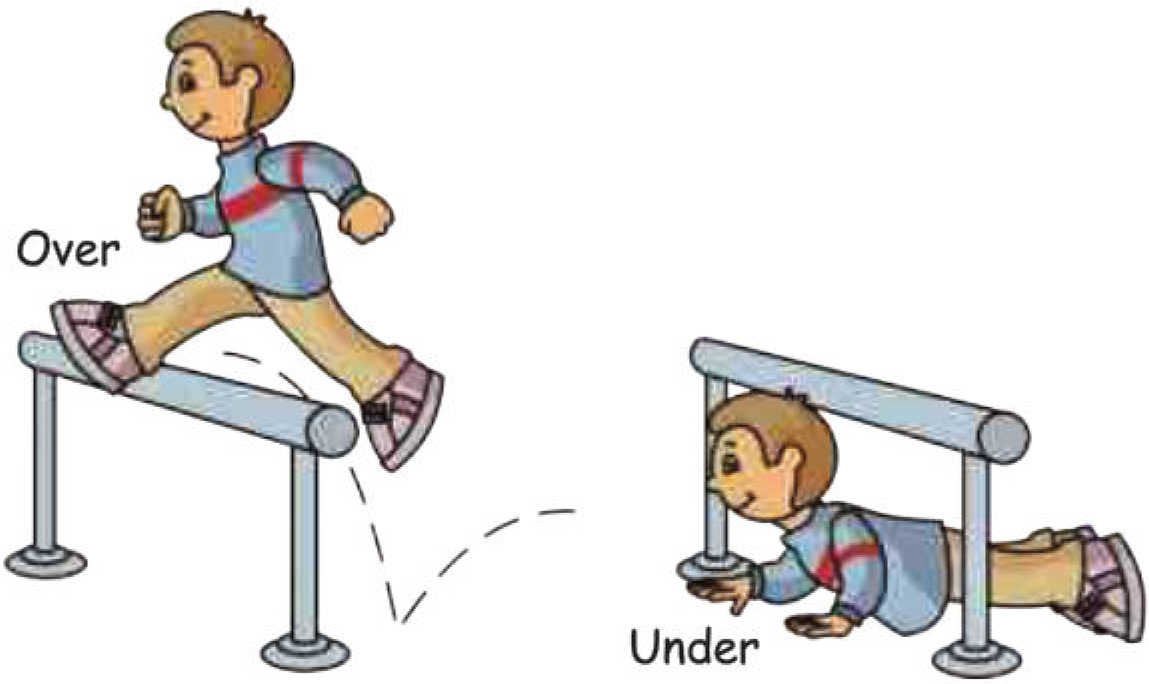
1. **Inside and Outside**

   The book is inside the bag and the register is outside the bag.

2. Above and below:

The picture is above the sofa and below the clock

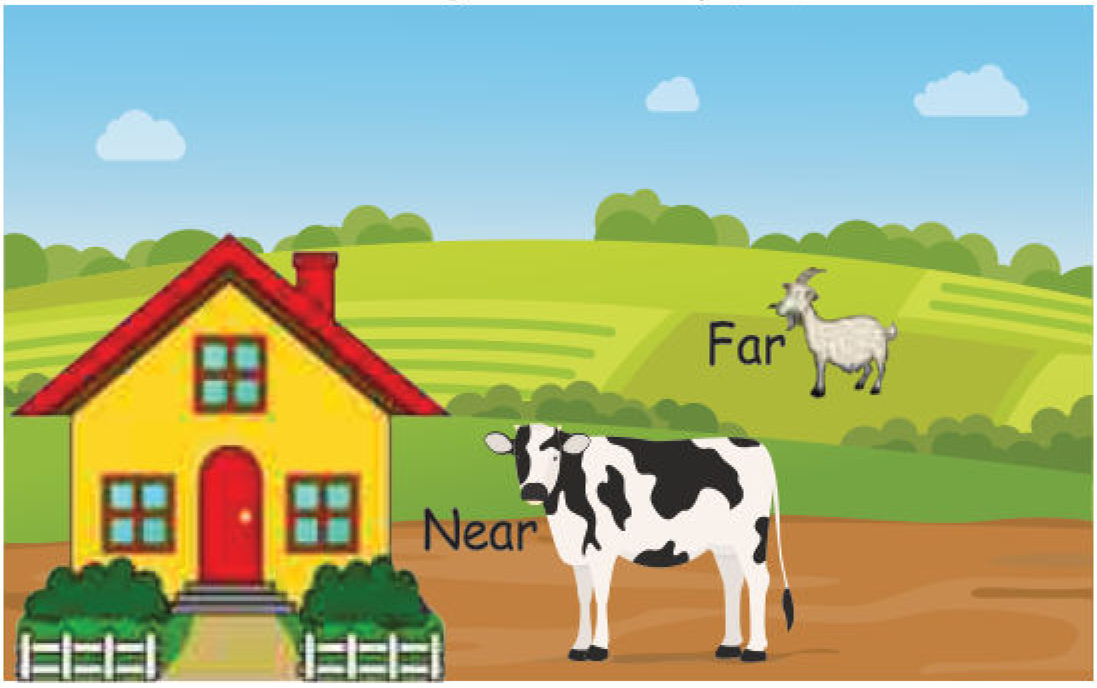
3. Over and Under:



The plane is flying over the city.

4. **Far and near:**

Cow is near the house and goat is far away from the house.



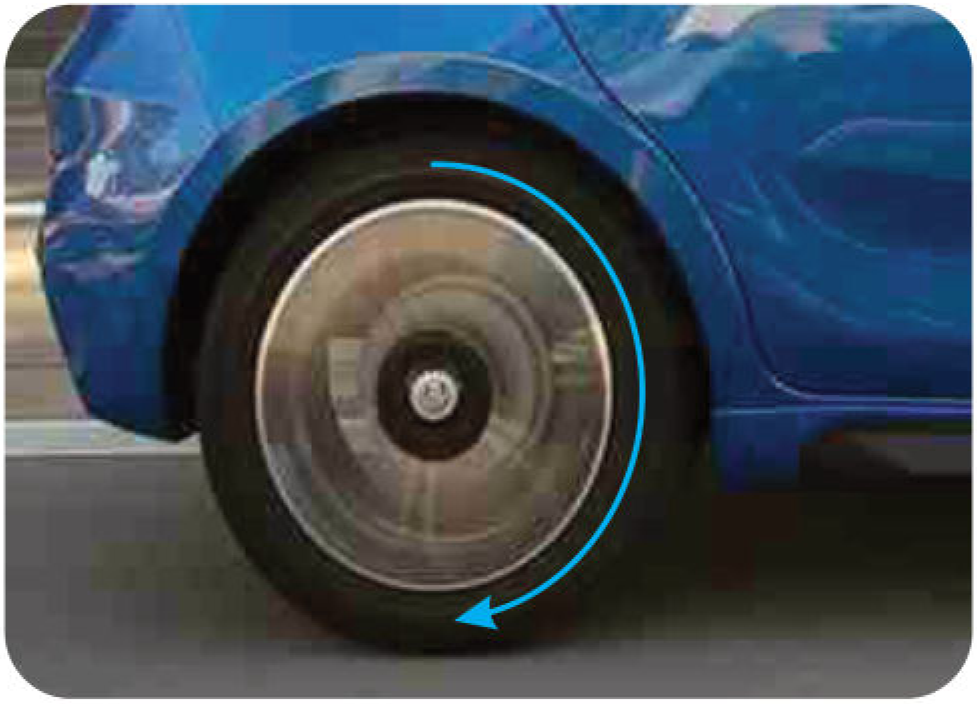
Describing Direction

When objects move, they can change their direction and orientation. We use specific words to describe these movements, including:

1. **Clockwise (turning right):**

A clock's hands moving from 12 to 3. This is a clockwise direction.

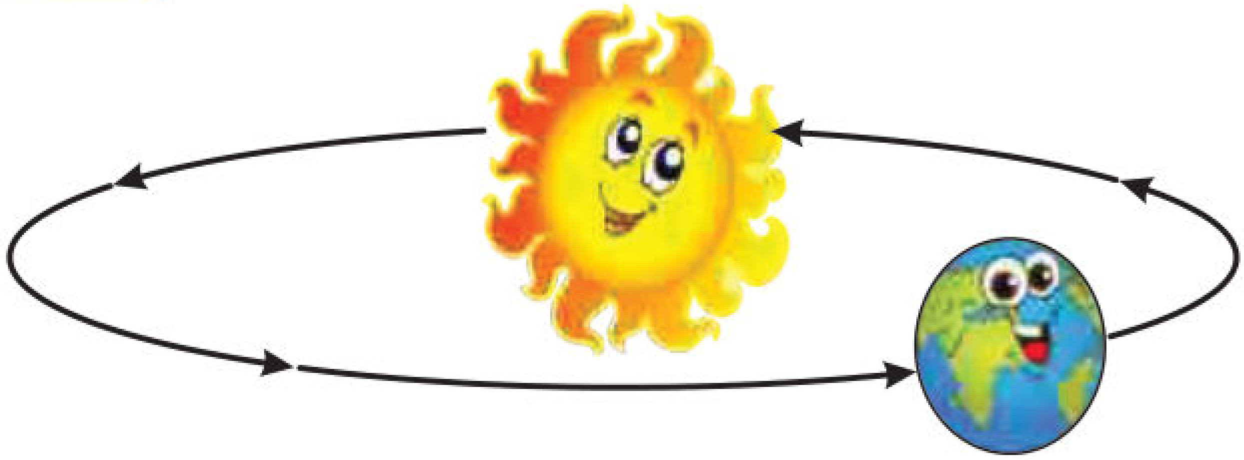
Example: The wheel is turning clockwise.



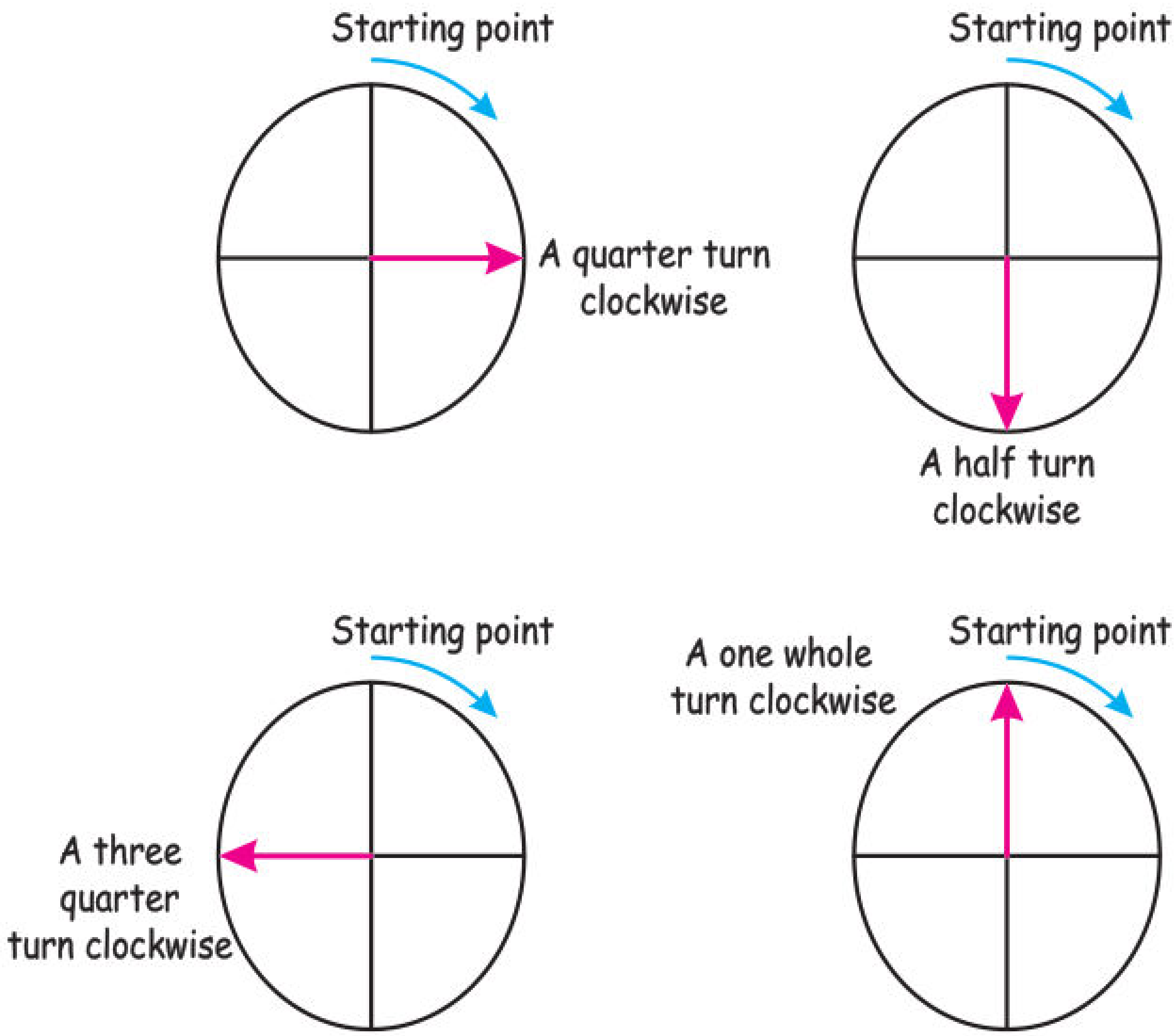
2. **Anti-clockwise (turning left):**

Imagine a clock's hands moving from 12 to 9. This is an anti-clockwise direction.

Example: The earth rotates around the Sun in anticlockwise direction.



3. Quarter turn, Half turn and Three quarter turn:



Describing Movement:

Objects can move from one place to another.

Example:

The ball rolled around the corner of the wall and into the street.

1) The bus went up the hill to the school.

2) The dog ran up the stairs to greet its owner.

3) The cyclist pedaled along the bike path to the park.

Unit 9

STATISTICS AND PROBABILITY

Student Learning Outcomes:

At the end of the chapters, Students will be able to:

- ✓ Read and interpret the data using pictographs and block graphs (including real life problems).
- ✓ Describe the likelihood that everyday events will occur, using mathematical language (impossible, less, likely, more likely, unlikely and certain).